

ISLEM KOBBI

PhD Researcher in AI for Robotics | Reinforcement Learning, Autonomous Systems, Computer Vision

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PROFESSIONAL SUMMARY

AI-for-robotics PhD researcher focused on reinforcement learning, autonomous systems, computer vision, robotic perception, motion planning, and control. Experienced in developing learning-based decision-making systems, road-scene perception pipelines, and robotics simulation workflows using Python, PyTorch, TensorFlow, OpenCV, ROS, Gazebo, Docker, Linux, and C/C++. Targeting AI-for-robotics roles involving intelligent autonomous agents, perception, planning, and real-world deployment.

TECHNICAL SKILLS

AI / Machine Learning: Reinforcement Learning, Deep Learning, Computer Vision, Perception, Decision-Making, Motion Planning, Multi-Agent Systems, Model Evaluation

Frameworks / Libraries: PyTorch, TensorFlow, Keras, Scikit-Learn, XGBoost, OpenCV, NumPy, Pandas, Pillow, SciPy

Robotics / Autonomy: ROS, Gazebo, Robot Pose Estimation, Control Systems, Autonomous Driving Simulation, UAV Systems, Mobile Robots

Programming / Tools: Python, C/C++, Java, Git, Linux, Docker, MATLAB / Simulink, HTML, CSS, JavaScript, Django, Node.js

PROFESSIONAL EXPERIENCE

PhD Researcher - Inria / Sorbonne Universite

Paris, France | Oct 2023 - Present

- Researching reinforcement learning methods for decision-making and motion planning in autonomous driving and intelligent-agent settings.
- Designing and testing simulation-based evaluation workflows for learning-based planning, with attention to robustness, planning quality, and safety-relevant scenarios.
- Developing reproducible AI experiments in Python and deep learning frameworks while connecting learning, control, and autonomous-systems validation.

R&D Deep Learning Intern - Astek Group

Boulogne-Billancourt, France | Feb 2024 - Aug 2024

- Developed multimodal deep learning approaches for traffic sign reconstruction, detection, and recognition in road-scene environments.
- Worked on computer vision and perception pipelines for autonomous driving and intelligent transportation systems using state-of-the-art architectures.
- Contributed to R&D workflows covering data preparation, model experimentation, visual recognition, reconstruction, and performance evaluation.

R&D AI / Control Intern - Laboratoire de Commande des Processus (LCP)

Algiers, Algeria | Jan 2023 - Jun 2023

- Optimized reinforcement learning and deep learning methods for pose estimation in mono-agent and multi-agent robotic systems.
- Built and tested aerial and mobile robot simulation environments using ROS and Gazebo, combining AI, robotics, and control concepts.

Robotics Intern - Centre de Developpement des Technologies Avancees (CDTA)

Algiers, Algeria | Jun 2022

- Assembled and brought up a hexarotor drone platform for robotics experimentation, hardware setup, system initialization, and UAV testing.

SELECTED PROJECTS

Reinforcement Learning for Autonomous Driving Decision-Making: Developed learning-based approaches for decision-making and motion planning in autonomous driving contexts; focused on autonomous agent behavior, planning quality, and simulation-based evaluation. Keywords: Reinforcement Learning, Motion Planning, Autonomous Driving, Intelligent Agents.

Robot Mobile Controlled by Computer Vision: Designed and built a multitask mobile robot able to solve a maze and collect/sort objects based on shape and color. Keywords: Robotics, Computer Vision, Perception, Autonomous Navigation, Control.

Traffic Sign Reconstruction, Detection, and Recognition: Built deep learning solutions for road-scene perception tasks in intelligent transportation systems. Keywords: Computer Vision, Deep Learning, Perception, Autonomous Driving.

Predictive Modeling with Neural Networks and Regression Algorithms: Built models for purchase, delay, restaurant demand, and loan propensity prediction; achieved first-place accuracy in several data science competitions. Keywords: Machine Learning, Deep Learning, Regression, Model Evaluation.

EDUCATION

PhD in Engineering Science - Sorbonne Universite, Paris, France | Oct 2023 - Present

Research focus: reinforcement learning, decision-making, motion planning, and autonomous driving.

Master 2 in Intelligent Systems Engineering - Sorbonne Universite, Paris, France | Sep 2023 - Sep 2024

Excellence scholarship. Coursework: advanced machine learning, advanced image and sound processing, learning for human-machine interaction, AI for robotics.

State Engineering Degree + Master 2 in Automatic Control - Ecole Nationale Polytechnique d Alger, Algeria | Sep 2018 - Jun 2023

Coursework: advanced control, robotics, optimization, optimal control, industrial computing, nonlinear systems, power electronics.

CERTIFICATIONS AND LANGUAGES

Certifications: Machine Learning Specialization - Coursera & DeepLearning.AI; TensorFlow for AI, Machine Learning, and Deep Learning - Coursera & DeepLearning.AI; Introduction to Data Science - Coursera & IBM.

Languages: Arabic - Native; French - Professional; English - Advanced (EF SET C2).